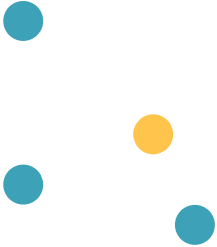


Wasserstein Barycenters

Barycenter: $\nu^\star = \arg \min_{\nu} \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$


$$\mathcal{B}_\lambda(x_1, \dots, x_4)$$

W barycenters as MM-OT

Wasserstein Barycenters

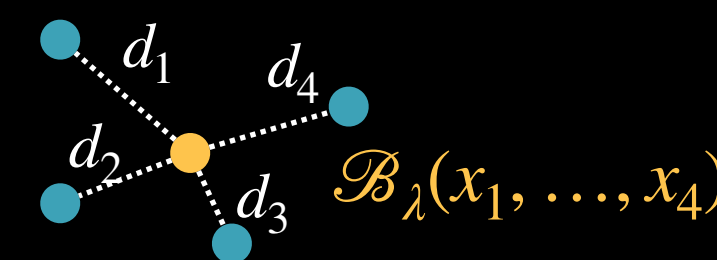
Barycenter: $\nu^\star = \arg \min_\nu \sum_{i=1}^K \lambda_i W_p^p(\mu_i, \nu)$

- Let S input measures $\mu_1, \dots, \mu_S$ with weights $\lambda_1, \dots, \lambda_S$, and barycenter measure be ν .

- Using gluing lemma, we can write the barycenter problem as:

$$\min_{\pi \in \mathcal{M}^{+1}(X^{S+1})} \int_{X^{S+1}} \sum_{s=1}^S \lambda_s c(x_s, x) d\pi(x_1, \dots, x_S, x) \quad \text{s.t. } (\text{Proj}_{x_s})\# \pi = \mu_s \quad \forall s \in \llbracket S \rrbracket.$$

- Here, barycenter plays role of $S + 1$ 'th marginal, with variable x
- Minimizing with respect to last marginal ν , one can show the cost can be written as an S -way cost as:

$$c(x_1, \dots, x_S) = \sum_{l=1}^S \lambda_l c(x_l, \mathcal{B}_\lambda(x_1, \dots, x_S)), \quad \text{where} \quad \mathcal{B}_\lambda(x_1, \dots, x_S) = \arg \min_x \sum_s \lambda_s c(x, x_s)$$


$c(x_1, \dots, x_4) = d_1 + \dots + d_4$

- Finding the barycenter amounts to solving an $(S\text{-way})$ MM-OT problem with this cost, and $\nu^\star = (\mathcal{B}_\lambda)\# \pi$

Multi-Marginal OT: Intuition

